

Module 2 Programming Part I



In-class Challenge Description:

In this exercise challenge, you will write a Java swing program that allows the user to enter personal information, including name and email address. The program should provide two main functions: submitting the entered information and displaying on the screen. This challenge tests your ability to create a functional GUI application.

Java Swing Project 1: User Information Form

Project Description: Create a Java Swing application for capturing user information. The application includes a form with various text input fields and buttons.

Key Elements:

1. A JFrame for the main application window.
2. Multiple JLabels for field descriptions.
3. Multiple JTextFields for user input.
4. At least two buttons, e.g., "Submit" and "Clear."

Functionality:

1. Users can input their information into the text fields.
2. "Submit" button: Validates and processes the user's input, e.g., displays it in a dialog box or saves it to a text file.
3. "Clear" button: Clears all text input fields.

Java Swing Project 2: Text File Reader/Writer

Project Description: Create a Java Swing application that allows users to read and write text files. The application will have a simple user interface with buttons for file operations.

Key Elements:

1. A JFrame for the main application window.
2. Two buttons: "Read File" and "Write File."
3. A JTextArea for displaying the content of the text file.
4. A JTextField for entering the file name.
5. JFileChooser for file selection.

Functionality:

1. "Read File" button: Opens a file dialog using JFileChooser, allows users to select a text file, reads the file's content, and displays it in the JTextArea.
2. "Write File" button: Opens a file dialog, lets users choose or create a text file, and writes the content from the JTextArea to the selected file.

Grading Rubric**Java Swing Project 2: Text File Reader/Writer (50 points)****1. User Interface (10 points):**

- The user interface is clear and organized.
- Proper use of buttons, text areas, and text fields.

2. Read File Functionality (20 points):

- Clicking "Read File" button reads the content of the specified text file.
- Displayed content is accurate and formatted.

3. Write File Functionality (20 points):

- Clicking "Write File" button writes the text area content to the specified text file.
- Successfully saves the content to the file.

4. Error Handling (10 points):

- Provides appropriate error messages for file-related issues.
- Gracefully handles exceptions.

Java Swing Project 1: User Information Form (50 points)

1. User Interface (10 points):

- The user interface is clear and well-organized.
- Proper use of labels, text fields, and buttons.

2. Input Validation (20 points):

- The "Submit" button correctly captures and displays user input.
- Handles empty input fields appropriately.

3. Display Functionality (10 points):

- Displays the user's information accurately.
- Information is properly formatted.

4. Clear Button Functionality (10 points):

- Clicking "Clear" button resets input fields and the result area.
- Successful reset without issues.

General Evaluation Criteria (20 points)

1. Code Quality (10 points):

- The code is well-structured and follows best practices.
- Proper indentation, meaningful variable names, and comments are used.

2. Ease of Use (10 points):

- The application is user-friendly and intuitive.
- Clear instructions for the user.

The total score for each project is 50 points, and the general evaluation criteria add up to 20 points, making a maximum total of 100 points for both projects. Grading should be based on the criteria specified in each section, and a total score is calculated for each project.

Points 100

Submitting a file upload

Due	For	Available from	Until
Sep 21	Everyone	-	Oct 22 at 11:59pm